

For this project, you'll be taking on the role of a financial planner, and will advise 3 different clients on how they should save and invest their money in their 401k based on their savings, age, risk tolerance, and expected yearly contribution. You will then compute the expected returns by the time these clients retire, how much money was put in to the account, and the amount earned through investing.

1 Information You Will Need

- The current retirement age in the US is 67 years old.
- Investments can be made in either **tax-advantaged** or **personal investment** accounts. Tax advantaged accounts (such as a 401k) do not have earnings taxed, so you will not need to consider taxes when money is being earned in them. However, contributions are capped. For a 401k, Personal contributions must be under \$19,500 per year, **not including employer match**. Including employer match, then maximum contribution is \$55,000 per year.
- There are hundreds of millions of things to invest in in the real world, but for this project you only have to consider the following potential investments.

Investment	Risk	Type	Expected Return (<i>compounded yearly</i>)
Apple	very high	stock, F500 company	25%
Google	very high	stock, F500 company	17%
Nasdaq	high	ETF (Tech stocks only)	15%
S&P500	medium-high	ETF (top 500 companies)	7%
Target date retirement fund	medium	ETF	5%
30 year Treasury bond	medium-low	Bond	3.5%
10 year Treasury bond	low	Bond	2.5%
High yield savings account	very low	savings account	1.5%

- You will need to consider inflation in your calculations. Expect that inflation will be around 2% per year, meaning that after you compute your annual return as a percent, you should subtract 2 from it due to inflation.
- Whenever anyone in this projects invests, they invest the entirety of their planned yearly contribution at the END of the year. This will be helpful in your calculations.

2 An Example

Alice is 40 years old. She currently has \$200,000 already in her 401k, and plans to maximize her contribution to her 401k every year until she retires, contributing at the beginning of the year, each year. For her 401k, she has an employer match up to \$10,000, meaning that every dollar she puts in to her 401k is matched by her employer, up to \$10,000. If she were to deposit any more money, she'd receive no match.

Alice is relatively risk averse. She wants 30% of her money in investments that are at most medium risk, 50% of her investments in investments that are at most medium-high risk, and the remaining 20% in high or very high risk investments.

Step 1: Compute Expected Returns (%)

You can decide to do this how you'd want, but I'll invest 30% of the money in the target date retirement fund, 50% in S&P 500, 10% in Apple, and 10% in Google. This satisfies Alice's requirements. To compute the rate of return before taxes, we take a weighted average:

$$\begin{aligned} & \underbrace{(0.30)}_{\% \text{ medium risk investment}} * \underbrace{(0.05)}_{\text{Rate of Return on target date fund}} \\ + & \underbrace{(0.50)}_{\% \text{ medium-high risk investment}} * \underbrace{(0.07)}_{\text{Rate of Return on S\&P500}} \\ + & \underbrace{(0.10)}_{\% \text{ Apple}} * \underbrace{(0.25)}_{\text{Rate of Return on Apple}} \\ + & \underbrace{(0.10)}_{\% \text{ Google}} * \underbrace{(0.17)}_{\text{Rate of Return on Google}} \end{aligned}$$

Giving us

$$0.3 * 0.05 + 0.5 * 0.07 + 0.1 * 0.25 + 0.1 * 0.17 = 0.092$$

Now we subtract 2% that goes to inflation, meaning that our net Expected Return is $0.092 - 0.02 = 0.072$, or 7.2%.

Step 2: Compute Gross Earnings on Initial Amount

First, we'll compute how much Alice will make with this investing strategy on the \$200,000 she already has in her 401k. She will invest for $67 - 40 = 27$ years at a rate of 7.2%, compounded yearly.

$$\$200,000(1 + 0.072)^{27} = \$1307040.96782 \approx \$1,307,030.97$$

Step 3: Compute Gross Earnings on Additional Investment

Alice will invest the maximum \$19,500 every year. Her employer will match the first \$10,000 she invests, so she will be investing $2 * (\$10,000) + \$9,500 = \$29,500$ every year. Thus, by investing \$29,500 per year at a 7.2% interest compounded yearly, with the money invested all at the beginning of the year, Alice will have the following amount in her account (from these investments) after the 27 years, when she retires:

$$\$29,500 \cdot \frac{(1 + 0.072)^{27} - 1}{0.072} \approx \$2,267,896.43$$

Step 4: Compute Total Amount in Account + Total amount invested (Principal)

When Alice retires, we expect that she will have $\$1,307,030.97 + \$2,267,896.43 = \$3,574,927.40$ in her account. To see how much money she put in, we add the \$200,000 that she already had to the \$19,500 she contributed every year for 27 years, meaning that she contributed

$$\$200,000 + 27 * \$19,500 = \underline{\$726,500}$$

To her investment account.

Step 5: Compute Net Earnings

Alice has \$3,574,927.40 in her account after investing \$726,500 of her own money, meaning she earned

$$\$3,574,927.40 - \$726,500 = \underline{\$2,848,427}$$

From her investments.

3 What You Need to Do

There will be a list of people just like Alice below. Choose any 3 you want, and do steps 1-5, as in the previous page, for them. I'll need the following numbers in your final report (with work shown!)

- Asset Allocation (i.e. what investments you recommend)
- Expected Returns as a percent
- Expected Gross Earnings (i.e. total amount in account on the year that the person retires)
- Total Amount Invested (i.e. the Principal)
- Expected Net Earnings

These numbers in the calculations above are underlined.

4 The People

- Bill is 27 years old and has \$20,000 currently in his 401k. He intends to contribute \$5,000 per year to his 401k every year until he retires. Bill gets no 401k matching from his employer. Bill is young, and prefers to invest in only ETFs and stocks. He wants 30% of his investments in stocks, and the remaining 70% in ETFs.
- Clara is 31 years old and has no money in her 401k at the moment. She feels bad about not investing earlier in her life, and now plans on investing the maximum amount into her 401k every year. Her employer offers her a \$5,000 401k match. Clara is a bit risk averse, but likes the upside of some high risk stocks. Clara wants 50% of her investments in medium or lower risk investments, 40% of her investments in ETFs, and 10% in stocks.
- Dave is 55 years old and has \$500,000 in his 401k at the moment. He plans on investing the maximum amount into his 401k every year, and gets a FULL (i.e. \$19,500) match from his employer every year. Dave is quite risk averse being so close to retirement, and wants 80% of his investments to be in bonds. The remaining 20%, however, can be in anything.
- Elia is 22 years old and has no money in her 401k. She just graduated college and is starting a new job that provides a \$2,000 401k match for the first 10 years, and a \$10,000 401k match for every year after that until she retires. Her job pays really well, and she plans on investing the maximum amount into her 401k every year. Elia wants 10% of her money to be in bonds, and the remaining 90% to be in ETFs.
- Farrah is 18 years old and not currently working. She was given \$10,000 from her parents to invest in her 401k. When Farrah starts working at 24, she plans on investing \$3,000 every year into her 401k. When she turns 40, she will up the amount to the full \$19,500. Farrah is assuming that her job will not provide any 401k matching. Farrah is very happy to take on risk, and wants to invest 50% into stocks and 50% in ETFs.
- Gerald is 50 years old and was recently promoted to a high level job. Gerald already has \$500,000 in his 401k and plans to contribute the maximum amount until he retires. With his promotion, he will get a whopping \$25,000 in employer matching. Gerald expects that when he turns 60, he will get promoted again, and his employer match will increase to \$40,000.
- As another option for this project, do this computation for yourself! Consider how old you are, what kind of 401k match you expect from your job (if you're unsure, \$5,000 is a reasonable number to work with), how much you plan on contributing to your 401k each year, and how risk averse you are. If you are considering this one, make sure to invest in at least 2 different things with your money.

5 Deadlines/Schedule

- 03/28: Receive project, pick people to advise.
- 03/30: Work on project in class, **rough draft due at the end of class.**
- 04/04: Work on project in class
- 04/06: **Present project in class** (less than 5 minutes)
- 04/11: **Written project due** to me by email **before class begins** at 2:00PM.

6 Grading

Along with attendance points, this project can earn you a maximum of 10 points. You will earn up to 3 points by presenting your work to the class, and up to 7 points from your written submission. Your rough draft submission will not be graded for correctness, but it is a valuable avenue to receive feedback on your work, and will be used to determine your daily points for the last week of March.

For extra credit, you can advise additional people. For every additional person you chose to advise, you'll earn up to an extra 3 points.